

PROFESSIONAL SUMMARY

Fresh B.Tech CSE graduate (DCRUST, 2026) with strong Python proficiency and hands-on experience in data collection, cleaning, and analysis using Pandas and NumPy — the core data preparation skills that underpin every AI/ML workflow. Independently built a complete end-to-end data pipeline demonstrating the same data engineering discipline required for ML model development: structured data collection, systematic cleaning, feature engineering, and validated analytical outputs. Solid understanding of ML/DL concepts including supervised and unsupervised learning, regression, classification, and clustering. Actively self-studying Scikit-learn to build practical modelling capability. Strong analytical mindset, willingness to learn rapidly, and genuine passion for AI/ML. Open to relocating to Noida upon receiving an offer.

TECHNICAL SKILLS

Python ★	Strong proficiency Pandas, NumPy — data manipulation, cleaning & transformation ETL scripting & automation Feature engineering Jupyter Notebook
Data Collection & Cleaning	Multi-source data gathering Missing value resolution Duplicate removal Type corrections Format standardisation Multi-stage validation
Feature Engineering	5 derived analytical features built from raw data: profit margin, age_group, transaction hour, shift classification, spending tier Rule-based & mathematical derivation
ML Concepts (Foundation)	Supervised learning — regression, classification Unsupervised — clustering Model evaluation concepts Train/test split logic Overfitting & underfitting understanding
Scikit-learn	Actively self-studying — no completed model yet Understanding of fit/predict workflow, preprocessing, pipelines Honest about current level
TensorFlow / PyTorch	Not yet used Deep learning concepts understood at theoretical level Actively exploring Will learn with guidance
SQL & Databases	MySQL, PostgreSQL Joins, CTEs, Window Functions 13 queries on real dataset Schema design & normalisation Data structures & DB concepts
Statistics	Descriptive statistics EDA Distribution analysis Correlation & trend analysis Segmentation
Data Visualisation	Power BI (4 live dashboards + DAX) Excel charts Analytical reporting

PROJECT EXPERIENCE

Retail Sales Analytics & Data Pipeline System | Python (Pandas, NumPy) · MySQL · Power BI · Excel | 2,000 transactions · 11 columns

Python — Data Collection, Cleaning & Transformation

- **Used Python (Pandas & NumPy)** as the primary tool for the full data preparation workflow — collecting raw records from Excel sources, cleaning inconsistencies, and transforming the dataset into a structured, analysis-ready format.
- **Resolved data quality issues** systematically — missing values filled or dropped based on field criticality, duplicate records removed, data types corrected, and date formats standardised — the same preprocessing discipline applied before feeding data into any ML model.
- **Automated the complete data pipeline** (Excel → Python → MySQL → Power BI) — scripted every step so the process runs consistently on any new data batch without manual intervention.

Feature Engineering — ML-Ready Data Preparation

- **Engineered 5 new analytical features** from raw data fields using Python — demonstrating the core data preparation skill that precedes every ML modelling task:
 - Profit Margin — mathematical derivation from revenue and cost fields
 - Age Group — rule-based categorical classification (Young / Adult / Middle Age / Senior)
 - Transaction Hour — temporal extraction from timestamp fields
 - Shift Classification — conditional binning into Morning / Afternoon / Evening
 - Spending Tier — threshold-based segmentation of transaction values

- **These features are exactly the type** of derived variables used to enrich datasets before training classification and segmentation ML models — turning raw fields into meaningful inputs that improve model performance.

Data Analysis, Validation & Statistical Insight

- **Conducted full EDA** across product, customer, and time dimensions — applying descriptive statistics, distribution analysis, correlation identification, and outlier detection to understand dataset structure before modelling.
- **Validated all analytical outputs** through multi-stage cross-checking — SQL null-check queries, Python type verification, and cross-layer reconciliation — mirroring the model testing and validation discipline required in ML workflows.
- **Identified 6 data-driven insights** from pattern and trend analysis — demonstrating the same analytical thinking applied to interpret model outputs and improve model performance:
 - Clothing = highest revenue → class imbalance signal in category models
 - Adults (25–40) = highest-value segment → key feature for customer classification models
 - Evening = peak period → time-based feature importance for demand prediction
 - Top-5 customers = ~15% revenue → outlier segment relevant to anomaly detection

AI/ML Concepts & Learning Trajectory

- **Solid conceptual foundation** in supervised learning (regression for value prediction, classification for category labelling), unsupervised learning (clustering for customer grouping), and model evaluation concepts (train/test split, accuracy, precision, recall).
- **Actively self-studying Scikit-learn** — understanding the fit/predict workflow, preprocessing pipelines, and common estimators. No completed production model yet — will be honest about this in any interview.
- **Coursework in Artificial Intelligence** from B.Tech CSE — covering search algorithms, neural network fundamentals, and optimization concepts — providing theoretical grounding for practical ML work.
- **Genuinely motivated to build ML capability** — the reason I am applying to Shurt Tech is to gain hands-on model development experience under guidance, not to claim experience I do not yet have.

EDUCATION

B.Tech — Computer Science & Engineering

2022 – 2026

Deenbandhu Chhotu Ram University of Science & Technology (DCRUST), Sonapat, Haryana

Relevant Coursework: Artificial Intelligence, Machine Learning Fundamentals, DBMS, Data Structures & Algorithms, Statistics & Probability, Python Programming, Software Engineering, Computer Networks

CORE COMPETENCIES

Python — Strong (Pandas · NumPy) • Data Collection & Cleaning • Feature Engineering • ML/DL Concepts — Foundation • Statistical Analysis & EDA • Data Validation & Testing • SQL & Database Concepts • ETL Pipeline Development • Analytical Problem-Solving • Data Structures & Algorithms • Scikit-learn — Actively Learning • Fast Learner • Growth Mindset